

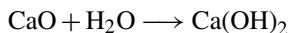
7.4.1 Limiting reagent calculations

For chemical reactions involving two or more reactants, it is necessary to determine which one is the limiting reagent. The following procedure is helpful in determining the limiting reagent:

1. Make sure the chemical equation is balanced.
2. Calculate the number of moles of each of the reactants present from their given amount.
3. Calculate the amount of product that can be formed from the complete reaction of each reactant.
4. Determine which of the reactants would produce the least amount of the product. This is the limiting reagent.

Example 7.8

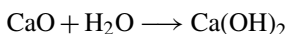
A mixture of 50 g of CaO and 50 g of H₂O react in an autoclave to form calcium hydroxide according to the equation



What is the limiting reagent?

Solution

1. We have a balanced equation.



2. Determine the number of moles of each reactant:

$$\text{Moles of CaO} = 50 \text{ g} \times \frac{1 \text{ mol CaO}}{56 \text{ g}} = 0.893 \text{ mol}$$

$$\text{Moles of H}_2\text{O} = 50 \text{ g} \times \frac{1 \text{ mol H}_2\text{O}}{18 \text{ g}} = 2.778 \text{ mol}$$

3. Calculate the moles of Ca(OH)₂ formed from each reactant:

Moles of Ca(OH)₂ produced from 0.893 mol CaO:

$$\text{Moles Ca(OH)}_2 = 0.893 \text{ mol CaO} \times \frac{1 \text{ mol Ca(OH)}_2}{1 \text{ mol CaO}} = 0.893 \text{ mol}$$

Moles of Ca(OH)₂ produced from 2.778 mol H₂O:

$$\text{Moles Ca(OH)}_2 = 2.778 \text{ mol H}_2\text{O} \times \frac{1 \text{ mol Ca(OH)}_2}{1 \text{ mol H}_2\text{O}} = 2.778 \text{ mol}$$

4. Now determine the limiting reagent. Since CaO produces the least amount of Ca(OH)₂, it is the limiting reagent.

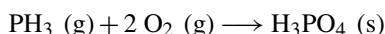
7.5 Reaction Yields: Theoretical, Actual, and Percent Yields

When reactions are carried out in the laboratory, the amount of product isolated is always less than the amount predicted from the balanced equation. There are several reasons for this. For example, some of the products may be lost during isolation and purification, some undesired products may be formed due to side reactions, or the reactant may not undergo complete reaction. The *theoretical yield* is the maximum amount of a product that can be formed as calculated from a chemical equation representing the reaction. The *actual yield* of a product is the amount that is actually formed when the experiment is performed. The *percent yield* is the ratio of the actual yield to the theoretical yield multiplied by 100. That is:

$$\text{Percent yield (\%)} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

Example 7.9

Phosphoric acid can be formed from the combustion of PH_3 in oxygen according to



If 188 g of phosphoric acid were produced in the reaction of 70 g of PH_3 in excess O_2 , calculate the theoretical yield and the percent yield.

Solution

1. We have a balanced equation.
2. Calculate the number of moles of PH_3 :

$$\text{Mol PH}_3 = 70 \text{ g} \times \frac{1 \text{ mol PH}_3}{33.97 \text{ g PH}_3} = 2.061 \text{ mol}$$

3. Determine the limiting reagent. Since oxygen is in excess, PH_3 is the limiting reagent.
4. Calculate the theoretical yield of H_3PO_4 using moles of the limiting reactant. Convert moles of H_3PO_4 to grams of H_3PO_4 :

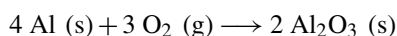
$$\text{g H}_3\text{PO}_4 = 2.061 \text{ mol PH}_3 \times \frac{1 \text{ mol H}_3\text{PO}_4}{1 \text{ mol PH}_3} \times \frac{98 \text{ g H}_3\text{PO}_4}{1 \text{ mol H}_3\text{PO}_4} = 201.98 \text{ g H}_3\text{PO}_4$$

5. Calculate the percent yield:

$$\text{Percent yield (\%)} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100 = \frac{188 \text{ g}}{201.98 \text{ g}} \times 100 = 93.1\%$$

Example 7.10

In a certain experiment, 70.0 g Al reacted with excess oxygen according to



If a yield of 82.5% was obtained, what was the actual yield of Al_2O_3 , in grams, from the experiment?

Solution

1. We have a balanced equation.
2. Calculate the number of moles of Al:

$$\text{Mol Al} = 75 \text{ g} \times \frac{1 \text{ mol Al}}{27 \text{ g Al}} = 2.778 \text{ mol}$$

3. Determine the limiting reagent. Since oxygen is in excess, Al is the limiting reagent.
4. Calculate the theoretical yield of Al_2O_3 using moles of the limiting reactant. Convert moles of Al_2O_3 to grams of Al_2O_3 :

$$\text{g Al}_2\text{O}_3 = 2.778 \text{ mol Al} \times \frac{2 \text{ mol Al}_2\text{O}_3}{4 \text{ mol Al}} \times \frac{102 \text{ g Al}_2\text{O}_3}{1 \text{ mol Al}_2\text{O}_3} = 141.68 \text{ g Al}_2\text{O}_3$$

5. Calculate the actual yield:

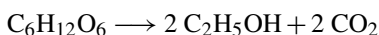
$$\text{Percent yield (\%)} = \frac{\text{actual yield}}{\text{theoretical yield}} \times 100$$

$$82.5\% = \frac{\text{actual yield}}{141.68} \times 100$$

$$\text{Actual yield} = \frac{141.68 \times 82.5}{100} = 117 \text{ g Al}_2\text{O}_3$$

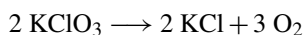
7.6 Problems

1. Consider the formation of ethanol by the fermentation of glucose, given by:



Calculate the maximum masses of ethanol and gaseous carbon dioxide that can be produced from the fermentation of 250 g of glucose.

2. What mass of oxygen will react with 10 g of hydrogen to form water?
3. The thermal decomposition of potassium chlorate is represented by:



Calculate:

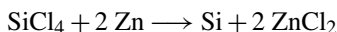
- (a) the mass of KClO_3 needed to generate 6.25 moles of oxygen.
- (b) the moles of KCl produced from the decomposition of 5.125 g of KClO_3 .
- (c) the mass and number of moles of oxygen produced from the decomposition of 5.125 g of KClO_3 .

4. Chlorine is prepared in the laboratory by the action of hydrochloric acid on manganese dioxide, MnO_2 , according to:



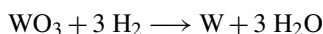
What mass of MnO_2 is required to completely react with 10.25 g of HCl ?

5. Very pure silicon (used in computer chips) is manufactured by heating silicon tetrachloride, SiCl_4 , with zinc:



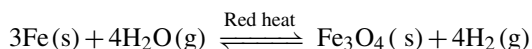
If 0.25 mol of Zn is added to excess SiCl_4 , how many grams of silicon are produced?

6. Tungsten metal, W , is used widely in the industrial production of incandescent bulb filaments. The metal is produced by the action of hydrogen on tungsten (VI) oxide:



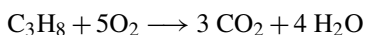
If a sample of WO_3 produces 25.825 g of water, how many grams of W are formed?

7. When steam is passed over iron filings at red heat, tri-iron tetroxide (Fe_3O_4) and hydrogen are produced. The balanced equation for the reaction is:

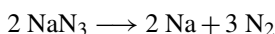


Calculate the volume of hydrogen produced at STP by reacting 0.25 mole of Fe with steam.

8. Calculate the volume of carbon dioxide produced at STP when 3.25 g propane burns in a rich supply of oxygen.

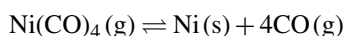


9. Large quantities of nitrogen gas can be produced from the thermal decomposition of sodium azide (NaN_3):



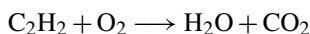
Calculate the volume of nitrogen produced at STP by the thermal decomposition of 85.25 g of NaN_3 .

10. Nickel tetracarbonyl, $\text{Ni}(\text{CO})_4$, decomposes at elevated temperature according to:



Calculate the volume of carbon monoxide produced at STP by the decomposition of 0.050 mol of $\text{Ni}(\text{CO})_4$.

11. Acetylene gas, which is used in welding, reacts with oxygen according to the equation:



(a) Balance the equation for the reaction.

(b) How many moles of oxygen are needed for the complete combustion of 5.25 g of acetylene?

(c) Calculate the volume at STP of carbon dioxide obtained from 5.25 g of acetylene.